

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Mid Term Examination 2025

Paper: Computer Architecture

Max Marks: 30

Time Allowed: 1 hour 15 mins

Q1) What is instruction set design. Discuss any two instruction set design with the help of example.

Q2) Describe the Following

a) Difference between RISC and CISC.

b) Flynn's taxonomy and describe any two types

Q3) Assume the stages of pipelined are perfectly balanced (ideal conditions), Calculate the time per instruction on the pipelined processor for both instructions a and b separately.

	Fetch	Decode	Execute	Memory	Writeback
a.	300ps	400ps	350ps	550ps	100ps
b.	200ps	150ps	100ps	190ps	140ps

Q # 4 what is cache memory? Discuss the significance of Cache memory

$$T_{total} = (n-1) \times T_{stage} + T_{writeback}$$

$$= (5-1) \times 100 + 100$$

$$= 500 + 100$$

$$= 600 \text{ ps}$$

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Final Term Examination 2025

Class: BSCS

Paper: Advance Database Systems

Student Name: _____

Semester: V

Max Marks: 50

Session (2023-2027)

Time Allowed: 3 Hours

Roll No: 095

Subjective Section (Marks = 35)

Q 1) Write short and precise answers.

(a) Write a MongoDB query to insert a new document into a students collection with fields: name, age, department, and GPA.

PLO	CLO	Bloom Taxonomy
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PLO-2	CLO-3	C3
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(b) Write a MongoDB query to delete all documents from the orders collection where the order status is "canceled".

PLO-2	CLO-3	C3
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(c) Write a MongoDB query to retrieve only the name and email fields for all customers in the customers collection.

PLO-2	CLO-3	C3
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(d) What is the difference between symmetric and asymmetric encryption?

PLO-2	CLO-2	C2
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Q 2) Write a PL/SQL block that accepts a department number and prints the total salary of all employees in that department. Use an aggregate function inside SELECT INTO. Handle NO_DATA_FOUND and OTHERS exceptions (10)

PLO-2	CLO-2	C3
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Q 3) Define ACID properties of a transaction in detail. Explain how each property ensures reliable and consistent database operations (5)

PLO-2	CLO-1	C2
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Q 4) Explain the importance of indexing in physical database design. Discuss different types of indexes (primary, secondary, clustered, non-clustered, composite, and bitmap) and how each affects query processing and performance tuning. (10)

PLO-2	CLO-2	C3
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Lahore College for Women University, Lahore

Class: BSCS

Semester: V

Course Title: Technical and Business Writing

Term: Final

Course code: ENG-384

Time allowed: 3 hrs

Total marks: 35+15=50

SUBJECTIVE TYPE

Attempt all Questions.

Q1. Describe the factors which influence the process of writing a technical report. 10 marks

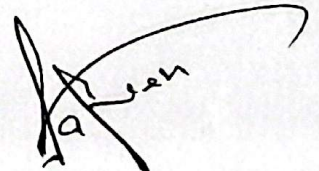
Q2. What is meant by a document design? Explain the main elements of designing a professional document. Discuss how well-designed documents improve readability and presentation. 15 marks

Q3. Write a brief note on the topics given below: 3+3+4

- i) Abstract
- ii) Citation
- iii) Project report

OBJECTIVE TYPE

- i. What is Bibliography and why is it important in report writing? 5 marks
- ii. Write three advantages of electronic document design. 5 marks
- iii. What is the purpose of an Abstract in a report? 5 marks



Lahore College for Women University, Lahore
Final Term Examination

Course Title: Database Systems

Maximum marks: 50 (Time: 3 Hrs)

Program: BSCS (IV)
Session (2023-2027)

4. List all products with their total quantity sold.
5. Find the most expensive product ordered.
6. Get customers who placed more than 3 orders.
7. Show orders with no items.
8. List top 3 customers by total spending.

Q4)

(4)

Considering the hierarchy of your university, employee can be of two types. Teaching & non-teaching. Draw separate EER segment for each of the following situations

- a) At a given time, an employee must be exactly one of these subtypes.
- b) An employee may or may not be one of these subtypes but at a given time, employee cannot be part of more than one subtype.
- c) An employee must be a member of one of these subtypes but at the same time an employee can perform any non-teaching task assigned by the management.

Ayesha

14/04/2023

Lahore College for Women University, Lahore
Final Term Examination

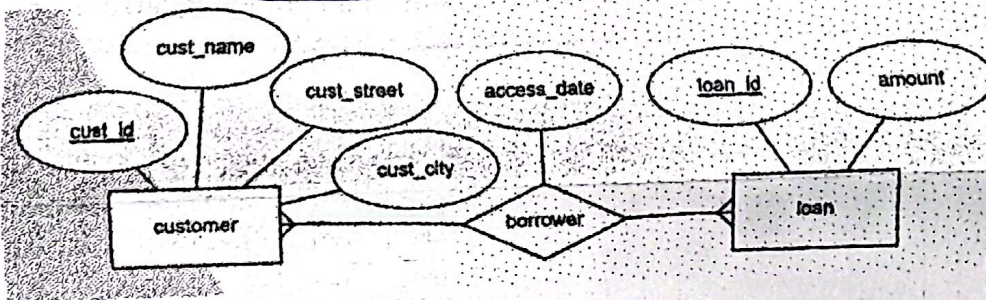
Course Title: Database Systems
Maximum marks: 50 (Time: 3 Hrs)
Q1)

Program: BSCS (IV)
Session (2023-2027)
(5+5)

a) Convert the following table into 3NF.

OrderProduct Table											
Order No	Prod No	Cust No	Name	Addr	City	St	Zip	Order Date	Promised Date	Desc	Qty Ord
61384	A128	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Bookcase	4
61384	B381	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Cabinet	2
61384	R210	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Table	1
62890	W891	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Chair	2
62890	A128	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Bookcase	8
63129	W891	1273	Cont. Designs	123 Oak	Austin	TX	78746	12/10/15	12/29/15	Chair	6
64000	B381	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Cabinet	1
64000	A128	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Bookcase	1

b) Convert the following ER diagram into Relation.



Q2) Define the following with one example of each concept.

(10)

- Derived Attribute
- Unary Relationship
- Associative Entity — *used to show Many-to-many relation*
- Completeness Constraints —
- Weak Entity

Q3) Write down the following SQL queries. Consider the following Schema.

(8*2)

Customers(customer_id, name, city)
Orders(order_id, customer_id, order_date, total_amount)
Products(product_id, name, price)
OrderItems(order_id, product_id, quantity)

- List all customers from 'New York'.
- Show each order with the customer name.
- Find total amount spent by each customer.

Agenda
14 Nov 2023

P-T-O

Q # 1 Write short answers of the following

1. How does pipelining improve the performance of CPU?
2. Define cache hit and cache miss?
3. Define Instruction format?
4. What is direct mapping of cache?
5. What is paging in virtual memory?

			(15)
CLO2	PLO1	C2	
CLO1	PLO1	C2	
CLO1	PLO1	C1	
CLO2	PLO2	C1	
CLO2	PLO2	C2	

Q # 2 Discuss in detail

1. Describe the need of I/O interface? Also discuss the interrupts in 8085 microprocessor?
2. What is multithreading? Discuss coarse-grained and fine-grained multithreading?
3. Differentiate between out-of-order execution and speculation in instruction level parallelism?
4. Differentiate between data stripping and data mirroring in RAID 0 and RAID 1 respectively?

			(20)
CLO3	PLO2	C2	
CLO3	PLO2	C2	
CLO3	PLO2	C2	
CLO3	PLO2	C2	

Q # 3 Analyze the following

1. A computer System in using demand paging with the following reference string 7,0,1,2,0,3,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Using the FIFO algorithm show step by step how pages will be loaded and calculate the number of page faults? (Assume three frames)

CLO2 PLO2 C2

2. A CPU has a clock cycle time of 2 ns and a five stage pipeline. Assume that each stage takes one clock cycle and there are 100 instructions to be executed

CLO2 PLO2 C2

- Calculate the total execution time with and without pipelining
- Determine the speedup ratio achieved due to pipelining

3. A cache memory has the following parameters:

CLO2 PLO2 C2

- a. Cache access time = 20ns, main memory access time = 100 ns, Hit ratio = 0.9

- b. Compare the effective memory access time (EMAT). Also, explain how increasing the cache size may effect this value.

Handwritten signature and date:
27-11-23

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE
Mid / Final Examination 2025

Class: BS-CS

Paper No: CS-425

Note: Attempts all questions.

Subject: Information Security

No. of credit hours: 3

Semester: V

Max Marks: 50

Term: Final

Time allowed: 3hrs

S.No	Questions	M	BT	CO	PO
Part -A(Short Answer Questions)					
1	What is meant by security attack? Explain various types of security attacks.	2	L1	CO1	PO1
2	Why do we need Diffie Hellman algorithm?	2	L2	CO2	PO3
3	Convert the Given Text "CRYPTOGRAPHY" into cipher text using Rail fence Technique. $k=3$	2	L3	CO1	PO1
4	What is the difference between a MAC and a one-way hash function	2	L2	CO2	PO3
5	Describe three ethical issues that should be considered when using computers.	2	L3	CO2	PO3
Part- B(Long Answer Questions)					
6	Use the Chinese Remainder Theorem to find the value of X that simultaneously solve following :- $2X \equiv 6 \pmod{14}$ $3X \equiv 9 \pmod{15}$ $5X \equiv 20 \pmod{60}$	5	L3	CO4	PO3
7	Consider a Diffie-Hellman scheme with prime $n = 11$ and a $g = 7$, then a. If user A pick his private key $XA = 6$, what is A's public key YA ? b. If user B pick his private key $XB = 9$, what is the secret key K shared with A?	5	L3	CO4	PO4
8	Differentiate Block cipher and Stream Cipher . Draw general depiction of DES Encryption Algorithm.	5	L4	CO2	PO3
9	Name the protocols that provide security in IPSec,	5	L2	CO4	PO5

***BloomsTaxonomyLevel(BT)**

(L1-Remembering;L2-Understanding;L3-Applying;L4-Analyzing;L5-Evaluating;L6-Creating)

Course Outcomes (CO), Program Outcomes (PO)

Flame

27-11-2025

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Mid Term Examination 2025

095

Paper: Computer Graphics

Max Marks: 30

Time Allowed: 1 hour 15 mins

1. Explain briefly components and working of CRT monitors. (5)
2. State the main difference between the midpoint circle and ellipse algorithms in terms of: symmetry handling, decision regions (single vs. two regions), and ending condition. (9)
3. Drive DDA algorithm for a straight line. (6)
4. Use the mid-point circle algorithm to draw the circle centered at (0,0) with radius 20. Draw the complete circle. (10)

